

CYK (Cocke - Younger - Kasami) Algorithm (or CKY algorithm)

It is one of the earliest recognition and parsing Algorithm.

It is used to solve membership problem using Dynamic Programming approach.

Membership Problem

Given a grammar G and a string w generates a Language $L(G)$.

Is the given string w a member of $L(G)$?

The algorithm requires the grammar G to be in Chomsky Normal Form (CNF).

This restriction is employed so that each problem can only be divided into two subproblems (not more - to bound time complexity).

It is bottom UP parsing algorithm. The parsing is a process to parse a string $w \in \Sigma^*$ (to find a derivation of w in the grammar G or to determine that there is none).

How does the CYK algorithm work?

For a string w of Length N , Construct a table T of size $N \times N$.

Each cell in the table $T[i,j]$ is the set of Constituents (Variables or Nonterminals) that can produce the substring spanning from position i to j . The process involves filling the table with the solutions to subproblems encountered in the bottom-up parsing process. Therefore, cells will be filled from **left to right** and **bottom to top**.

The algorithm considers every possible subsequence of letters ~~starting from i to j~~ ~~and adds~~ non-terminal N to $T[i,j]$, if the sequence of letters starting from i to j can be generated from the non-terminal N .

For Subsequence of length 2 and greater,

it considers every possible partition of the subsequence into two parts, and checks if there is a rule of the form $A \rightarrow BC$ in the grammar where B and C can generate the two parts respectively, based on already existing entries in table T .

The Sentence ^{or string} can be produced by the grammar only if the entire string is matched by the start symbol i.e., if S is a member of $T[1,n]$.

Consider the follow CFG in Chomsky Normal Form:

$$S \rightarrow AB \mid BC$$

$$A \rightarrow BA \mid a$$

$$B \rightarrow CC \mid b$$

$$C \rightarrow AB \mid a$$

CYK algorithm is used to find whether the given grammar generates a string **baaba** or not.

As $w = baaba$ having $|w| = 5$, so a table of size 5×5 is needed.

		b	a	a	b	a
		1	2	3	4	5
b	1	{B}	{A, S}	$\emptyset$	$\emptyset$	{S, A, C}
a	2		{A, C}	{B}	{B}	{S, A, C}
a	3			{A, C}	{S, C}	{B}
b	4				{B}	{A, S}
a	5					{A, C}

Step I: Subsequena of length 1 ($w = \overset{w_1 w_2 w_3 w_4 w_5}{baaba}$).

L to R and Bottom to Top

Compute enterkes for $T[1,1]$, $T[2,2]$, $T[3,3]$, $T[4,4]$ and $T[5,5]$.

2-Dimensional representation

$w = \overset{1 \ 2 \ 3 \ 4 \ 5}{b \ a \ a \ b \ a}$

$w_1 = b$

$T[1,1] = ?$ $1 = 11 = b$

find the rule of the form $N \rightarrow b$

$B \rightarrow b$ place B to cell $T[1,1]$.

$$S \rightarrow AB \mid BC$$

$$A \rightarrow BA \mid a$$

$$B \rightarrow CC \mid b$$

$$C \rightarrow AB \mid a$$

$T[2,2] = ?$ $w = \overset{1 \ 2 \ 3 \ 4 \ 5}{b \ a \ a \ b \ a}$

2-Dimensional representation of 2 $2 \text{ or } 22 = a$

find a rule of the form $N \rightarrow a$

$$A \rightarrow a$$

$$C \rightarrow a$$

Place A, C to cell $T[2,2]$

$$S \rightarrow AB \mid BC$$

$$A \rightarrow BA \mid a$$

$$B \rightarrow CC \mid b$$

$$C \rightarrow AB \mid a$$

$T(3,3):$

$w = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$

3 or 33 = a

find a rule of the form $N \rightarrow a$
we have, two rules

$A \rightarrow a$

$C \rightarrow a$

Place $\{A, C\}$ in $T(3,3)$

$T(4,4):$

$w = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$

4 or 44 = b

there is one rule of the form $N \rightarrow b$

$B \rightarrow b$

Place in Cell $T(4,4)$.

$T(5,5):$

$w = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$

5 or 55 = a

$A \rightarrow a$

$C \rightarrow a$

Place $\{A, C\}$ in $T(5,5)$

STEP 2

This completes the bottom (base case) level (Diagonal of Table T)

Now Compute 2nd diagonal in bottom up fashion i.e;

1. $T(1,2) = ?$
2. $T(2,3) = ?$ one less than the above entries
3. $T(3,4) = ?$
4. $T(4,5) = ?$

1. $T(1,2) =$

$w = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$

$(1,2) = (1,1) \cdot (2,2)$

$= \{B\} \cdot \{A, C\}$

$= BA, BC$

fetch the entries from Table we have already calculated.

find the rules of the form $N \rightarrow BA$
 $N \rightarrow BC$

$= BA, BC$

$\uparrow \quad \uparrow$
 $A \quad S$

we have two rules

$A \rightarrow BA$

$S \rightarrow BC$

place $\{A, S\}$ to cell $T(1,2)$

$S \rightarrow AB \mid BC$

$A \rightarrow BA \mid a$

$B \rightarrow CC \mid b$

$C \rightarrow AB \mid a$

2. $T[2,3] =$ $\omega = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$

$$(2,3) = (2,2) \cdot (3,3)$$

$$= \{A, c\} \cdot \{A, c\} \quad \text{from table}$$

$$= AA, AC, CA, CC$$

$\begin{matrix} & \times & \times & \times & \uparrow \\ & & & & B \end{matrix}$

$$S \rightarrow AB \mid BC$$

$$A \rightarrow BA \mid a$$

$$B \rightarrow CC \mid b$$

$$C \rightarrow AB \mid a$$

$B \rightarrow CC$ Place B to cell $T[2,3]$

3. $T[3,4]:$

$\omega = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$ 3 | 4

$$(3,4) = (3,3) \cdot (4,4)$$

$$= \{A, c\} \cdot \{B\}$$

$$= AB, CB$$

$\begin{matrix} \uparrow \uparrow & \times \\ C & S \end{matrix}$

$$S \rightarrow AB \mid BC$$

$$A \rightarrow BA \mid a$$

$$B \rightarrow CC \mid b$$

$$C \rightarrow AB \mid a$$

As $S \rightarrow AB$
 $C \rightarrow AB$ Place $\{S, C\}$ in $T[3,4]$.

4. $T[4,5]:$

$\omega = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$ 4 | 5

$$(4,5) = (4,4) \cdot (5,5)$$

$$= \{B\} \cdot \{A, c\}$$

$$= BA, BC$$

$\begin{matrix} \uparrow & \uparrow \\ A & S \end{matrix}$

$$S \rightarrow AB \mid BC$$

$$A \rightarrow BA \mid a$$

$$B \rightarrow CC \mid b$$

$$C \rightarrow AB \mid a$$

As $A \rightarrow BA$
 $S \rightarrow BC$ Place $\{A, S\}$ to $T[4,5]$.

STEP 3

This completes Diagonal 2.

Now Compute 3rd diagonal in bottom-up way:

1. $T[1,3] = ?$

2. $T[2,4] = ?$

3. $T[3,5] = ?$

$T[1,3]:$

$\omega = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$

$$(1,3) = 1 \mid 2 \ 3 \quad \textcircled{1} \text{ two ways to decompose}$$

$$= 1 \ 2 \mid 3 \quad \textcircled{2}$$

① implies

$$1 \mid 2 \ 3 = (1,1) \cdot (2,3)$$

$$= \{B\} \cdot \{B\}$$

$$= BB$$

$\times$

② implies

$$1 \ 2 \mid 3 = (1,2) \cdot (3,3)$$

$$= \{A, S\} \cdot \{A, c\}$$

$$= AA, AC, SA, SC$$

$\begin{matrix} \times & \times & \times & \times \end{matrix}$

No rule here, so place ϕ in $T[1,3]$

$$T[2,4]: \quad w = b \overset{1}{a} \overset{2}{a} \overset{3}{b} \overset{4}{a}$$

$(2,4)$ means 2 to 4.

$$\begin{aligned} (2,4) &= 2 | 3 \ 4 = (2,2) \cdot (3,4) = \{A, C\} \cdot \{S, C\} = \overset{\times}{AS}, \overset{\times}{AC}, \overset{\times}{CS}, \overset{\uparrow B}{CC} \\ &= 2 \ 3 | 4 = (2,3) \cdot (4,4) = \{B\} \cdot \{B\} = \overset{\times}{BB} \end{aligned}$$

There is only one rule $B \rightarrow CC$, so place B in $T[2,4]$.

$$T[3,5]: \quad w = b \overset{1}{a} \overset{2}{a} \overset{3}{b} \overset{4}{a} \overset{5}{a}$$

$(3,5)$ means 3 to 5

$$\begin{aligned} (3,5) &= 3 | 4 \ 5 = (3,3) \cdot (4,5) = \{A, C\} \cdot \{A, S\} = \overset{\times}{AA}, \overset{\times}{AS}, \overset{\times}{CA}, \overset{\times}{CS} \\ &= 3 \ 4 | 5 = (3,4) \cdot (5,5) = \{S, C\} \cdot \{A, C\} = \overset{\times}{SA}, \overset{\times}{SC}, \overset{\times}{CA}, \overset{\uparrow B}{CC} \end{aligned}$$

There is only one rule $B \rightarrow CC$, so place B in $T[3,5]$.

This completes the 3rd diagonal

Step 4

Now Compute 4th diagonal.

$$T[1,4] = ? \rightarrow$$

$$w = b \overset{1}{a} \overset{2}{a} \overset{3}{b} \overset{4}{a}$$

$$T[2,5] = ?$$

$$\begin{aligned} (1,4) &= 1 | 2 \ 3 \ 4 = (1,1) \cdot (2,4) = \{B\} \cdot \{B\} = BB \\ &= 1 \ 2 | 3 \ 4 = (1,2) \cdot (3,4) = \{A, S\} \cdot \{S, C\} = \overset{\times}{AS}, \overset{\times}{AC}, \overset{\times}{SS}, \overset{\times}{SC} \\ &= 1 \ 2 \ 3 | 4 = (1,3) \cdot (4,4) = \emptyset \cdot \{B\} = \emptyset \end{aligned}$$

Place $\emptyset$ in $T[1,4]$

Concatenation of any set with empty set is empty set.

$$T[2,5] = ?$$

$$w = b \overset{1}{a} \overset{2}{a} \overset{3}{b} \overset{4}{a} \overset{5}{a}$$

$$\begin{aligned} (2,5) &= 2 | 3 \ 4 \ 5 = (2,2) \cdot (3,5) = \{A, C\} \cdot \{B\} = \overset{\uparrow S}{AB}, \overset{\uparrow C}{CB} \\ &= 2 \ 3 | 4 \ 5 = (2,3) \cdot (4,5) = \{B\} \cdot \{A, S\} = \overset{\uparrow A}{BA}, \overset{\uparrow S}{BS} \\ &= 2 \ 3 \ 4 | 5 = (2,4) \cdot (5,5) = \{B\} \cdot \{A, C\} = \overset{\uparrow A}{BA}, \overset{\uparrow S}{BC} \\ &= \cancel{2 \ 3 \ 4 \ 5} \cdot (1,4) \cdot (5,5) \end{aligned}$$

There are 3 rules :

$$S \rightarrow AB$$

$$C \rightarrow B$$

$$A \rightarrow BA$$

Place $\{S, A, C\}$ in $T[2,5]$.

This completes diagonal 4.

STEP 5:

Now Compute diagonal 5.

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$$T[1,5] = T[1,n] = ?$$

$$w = \overset{1}{b} \overset{2}{a} \overset{3}{a} \overset{4}{b} \overset{5}{a}$$

$$(1,5) = 1 | 2 \ 3 \ 4 \ 5 = (1,1) \cdot (2,5) = \{B\} \cdot \{S, A, C\} = BS, BA, BC$$

$\times \uparrow_A \uparrow_S$

$$= 1 \ 2 | 3 \ 4 \ 5 = (1,2) \cdot (3,5) = \{A, S\} \cdot \{B\} = AB, SB$$

$\uparrow_S \uparrow_C \times$

$$= 1 \ 2 \ 3 | 4 \ 5 = (1,3) \cdot (4,5) = \phi \cdot \{A, S\} = \phi$$

$$= 1 \ 2 \ 3 \ 4 | 5 = (1,4) \cdot (5,5) = \phi \cdot \{A, C\} = \phi$$

There 3 rules of the form

$$S \rightarrow BC$$

$$S \rightarrow AB$$

$$C \rightarrow AB$$

$$A \rightarrow BA$$

place S, C, A in $T[1,5]$.

The final table becomes

	$\overset{1}{b}$	$\overset{2}{a}$	$\overset{3}{a}$	$\overset{4}{b}$	$\overset{5}{a}$
$\overset{1}{B}$	B	A, S	ϕ	ϕ	S, A, C
$\overset{2}{A}$		A, C	B	B	S, A, C
$\overset{3}{A}$			A, C	S, C	B
$\overset{4}{B}$				B	A, S
$\overset{5}{A}$					A, C

~~Is~~ Is $baaba$ in $L(G)$?

We can see the S (start Nonterminal) in the set $T[1,n]$ or $T[1,5]$.

$$T[1,5] = S, A, C \text{ then}$$

$$baaba \in L(G) \text{ (As } S \in T[1,5]).$$

Time Complexity:

The running time of this algorithm is $O(n^3)$.

Theorem:

The CYK algorithm correctly computes $T[i,j]$ for all i and j ;
 w is in $L(G)$ if and only if S is in $T[1,n]$.